SYSTEM SPECIFICATION DOCUMENT

CEMU–SNW Social & Savings Association Management System

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1. SYSTEM OVERVIEW

1.1 System Name

CEMU–SNW Social and Savings Management System

Proposed link: https://cemu.snwgi.rw

1.2 Purpose

To automate and manage:

- Social Fund (cash book)
- Member Savings
- Loan processing and tracking
- Fund administration and approvals

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2. USER AUTHENTICATION MODULE

2.1 Login Gateway

Fields:

- Username (Staff ID)
- Password

Features:

- Secure login authentication
- Role-based access control

2.2 Forgot Password

- Reset via email or admin approval
- Security questions (optional)

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3. USER ROLES & ACCESS CONTROL

Role	Permissions
Member	View savings, loans, contributions, request to change contribution and request loan
Accountant	Manage transactions, reports
Loan Officers	Approve/reject loans
President	Final approval authority
Admin	Full system control

⚠️ **Fund Administration Module Access:**

- Restricted to:
  - Accountant (Executor)
  - President (Approver)
  - Loan Officer (Approver)

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4. DASHBOARD (HOME SCREEN)

4.1 Summary Widgets

Each displayed in separate panels:

Savings Summary

- Initial Capital: 50,000 FRW
- Cumulative Savings
- Cumulative Interest
- Total Savings

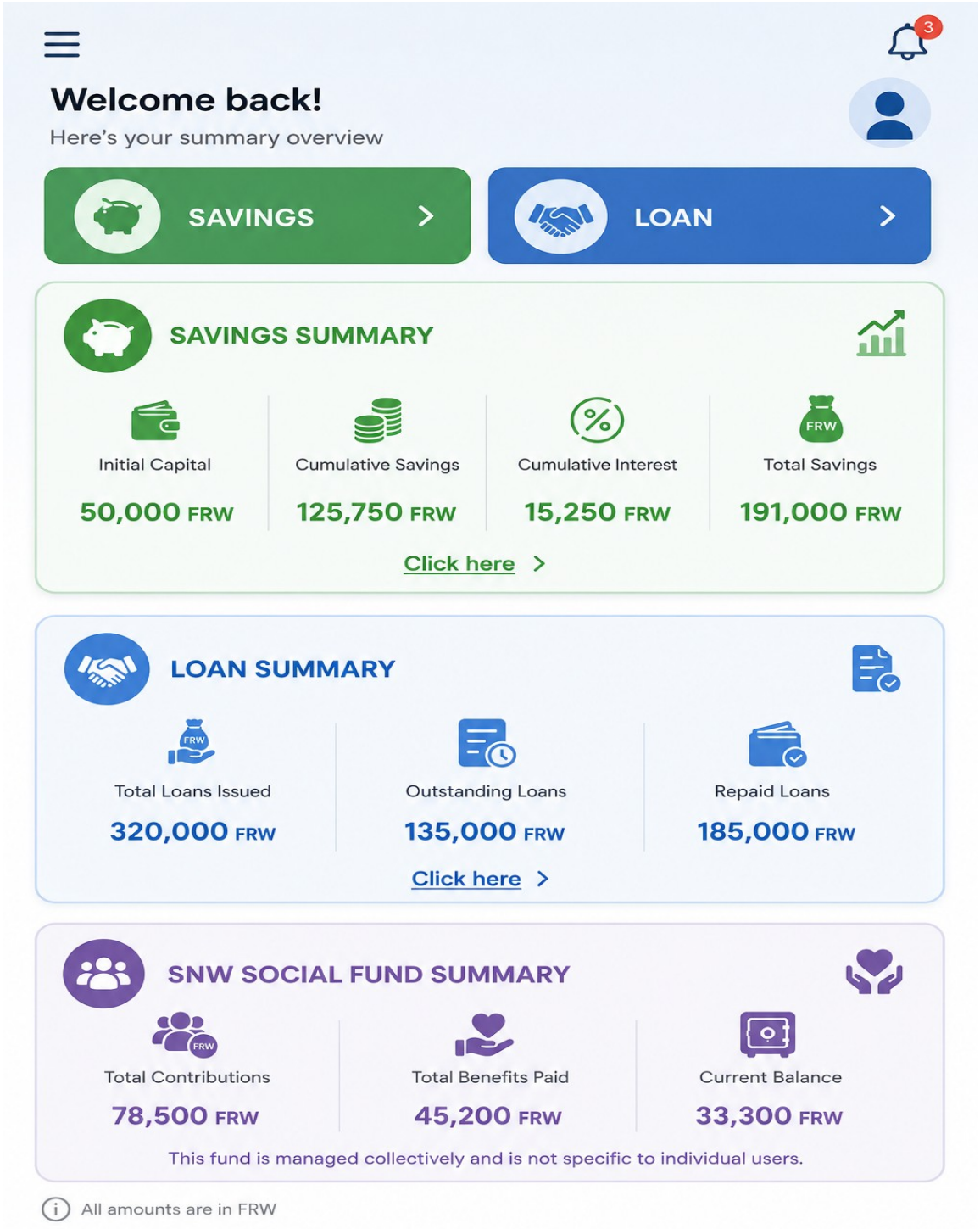
Loan Summary

- Total Loans Issued
- Outstanding Loans
- Repaid Loans

Social Fund Summary

- Total Contributions
- Total Benefits Paid
- Current Balance

Below is example: at first login face:


3
Welcome back!
Here's your summary overview
SAVINGS
LOAN
SAVINGS SUMMARY
Initial Capital
50,000 FRW
Cumulative Savings
125,750 FRW
Cumulative Interest
15,250 FRW
Total Savings
191,000 FRW
Click here
LOAN SUMMARY
Total Loans Issued
320,000 FRW
Outstanding Loans
135,000 FRW
Repaid Loans
185,000 FRW
Click here
SNW SOCIAL FUND SUMMARY
Total Contributions
78,500 FRW
Total Benefits Paid
45,200 FRW
Current Balance
33,300 FRW
This fund is managed collectively and is not specific to individual users.
All amounts are in FRW

5. SYSTEM MODULES

MODULE 1: SOCIAL FUND (CASH BOOK)

5.1 Features

- Monthly contributions tracking
- Benefit payments tracking
- Running balance calculation

5.2 Data Fields

Field	Description
Date	Transaction date
Description	Contribution or payment narration
Increase	Contributions
Decrease	Benefits paid
Balance	Running total

5.3 Transactions

Contributions

- Monthly: 10,000 FRW per member per payroll deduction
- Add contribution (specify period ,date, and amount) by accountant or upload excel

Benefits

Pay the benefit (choice circumstance) and enter the amount and narrations

- ICU Support: 1,000,000 FRW
- Death Benefit: 500,000 FRW
- Others: configurable

SOCIAL FUND ACCOUNT

Manage contributions and benefit payments

Export to Excel

+ Add New Contribution | Pay Benefit

DATE	DESCRIPTION	INCREASE (FRW)	DECREASE (FRW)	BALANCE (FRW)
01/01/2026	January 2026 contribution	600,000	–	600,000
01/02/2026	February 2026 contribution	600,000	–	1,200,000
01/03/2026	March 2026 contribution	600,000	–	1,800,000
01/04/2026	April 2026 contribution	600,000	–	2,400,000
05/05/2026	PAYMENT KAMANA XXXXX	–	1,000,000	1,400,000

CURRENT BALANCE
1,400,000 FRW

TOTAL CONTRIBUTIONS
2,400,000 FRW

TOTAL BENEFITS PAID
1,000,000 FRW

All amounts are in FRW

MODULE 2: SAVINGS MANAGEMENT

6.1 Features

- Individual member accounts
- Monthly contribution tracking
- Interest calculation
- Withdrawal management

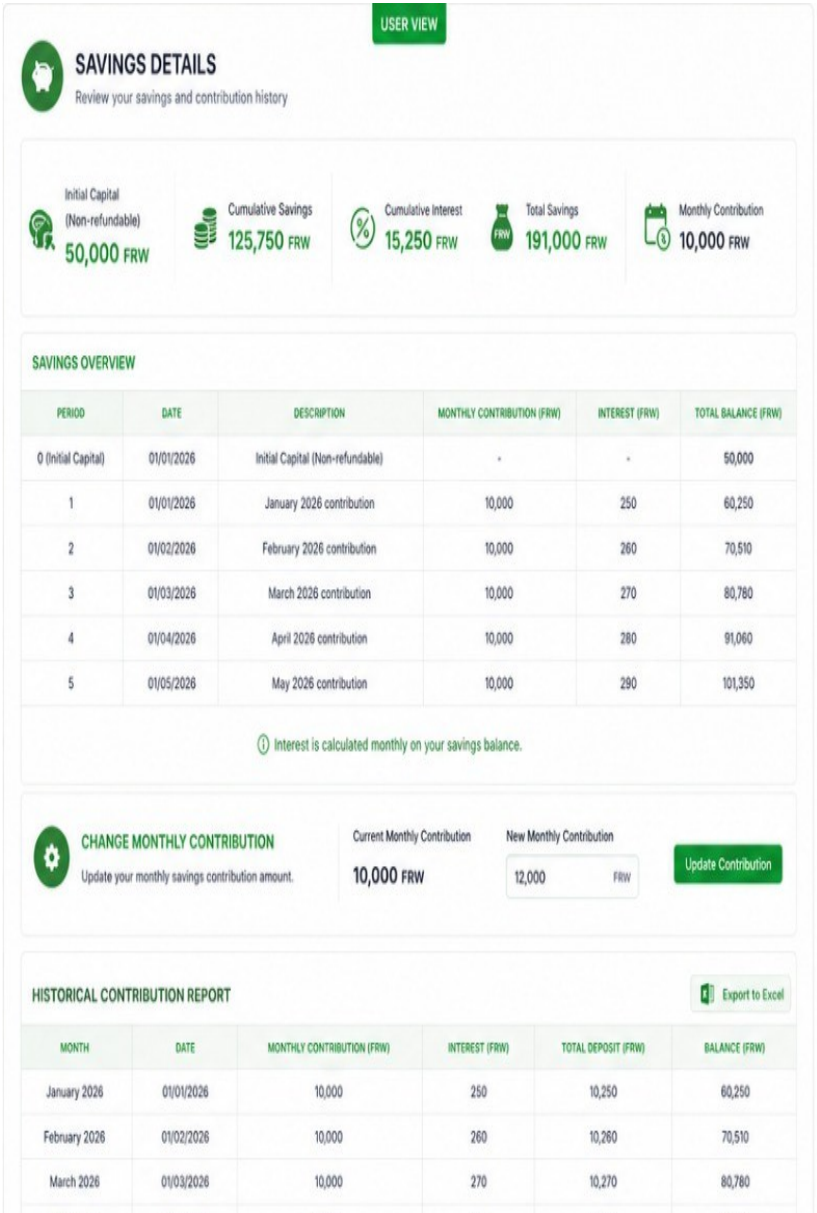
6.2 Data Structure

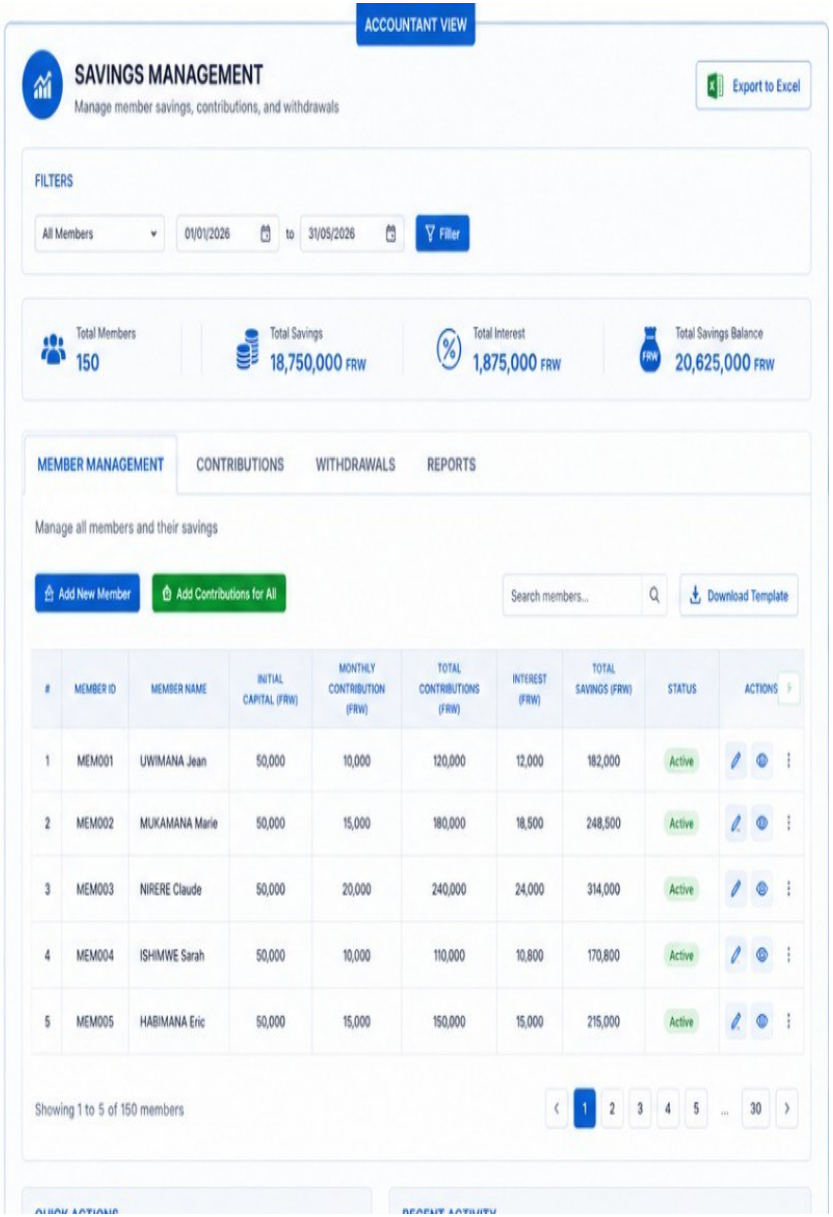
Field	Description
Period	Month/Year
Monthly Contribution	Amount saved
Interest Earned	Calculated interest
Cumulative Saving	Running total

6.3 Rules

- Minimum monthly saving: 15,000 FRW
- Share capital: 50,000 FRW (once)
- Savings withdrawal allowed only if:
    - No outstanding loan

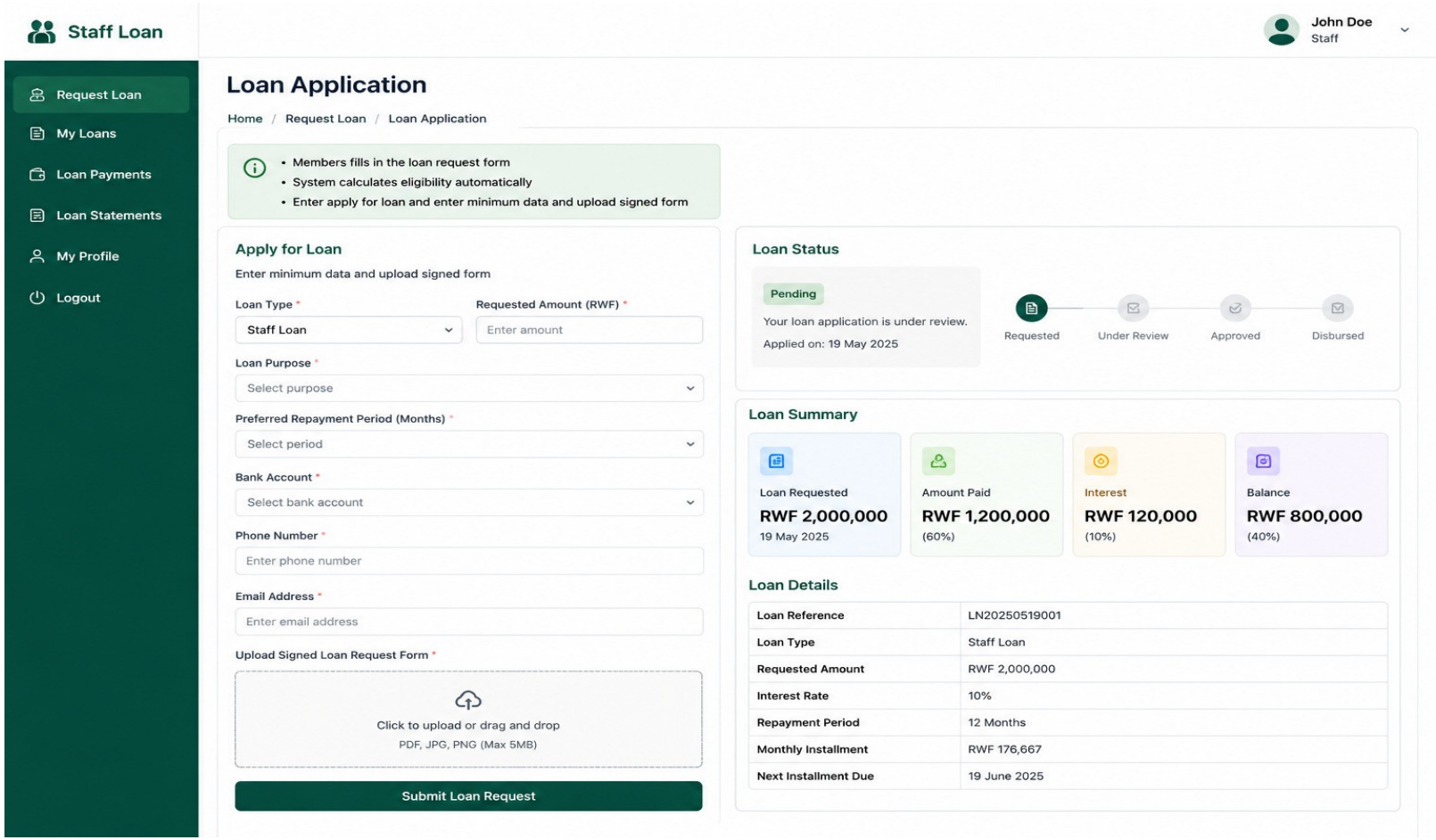
Example:

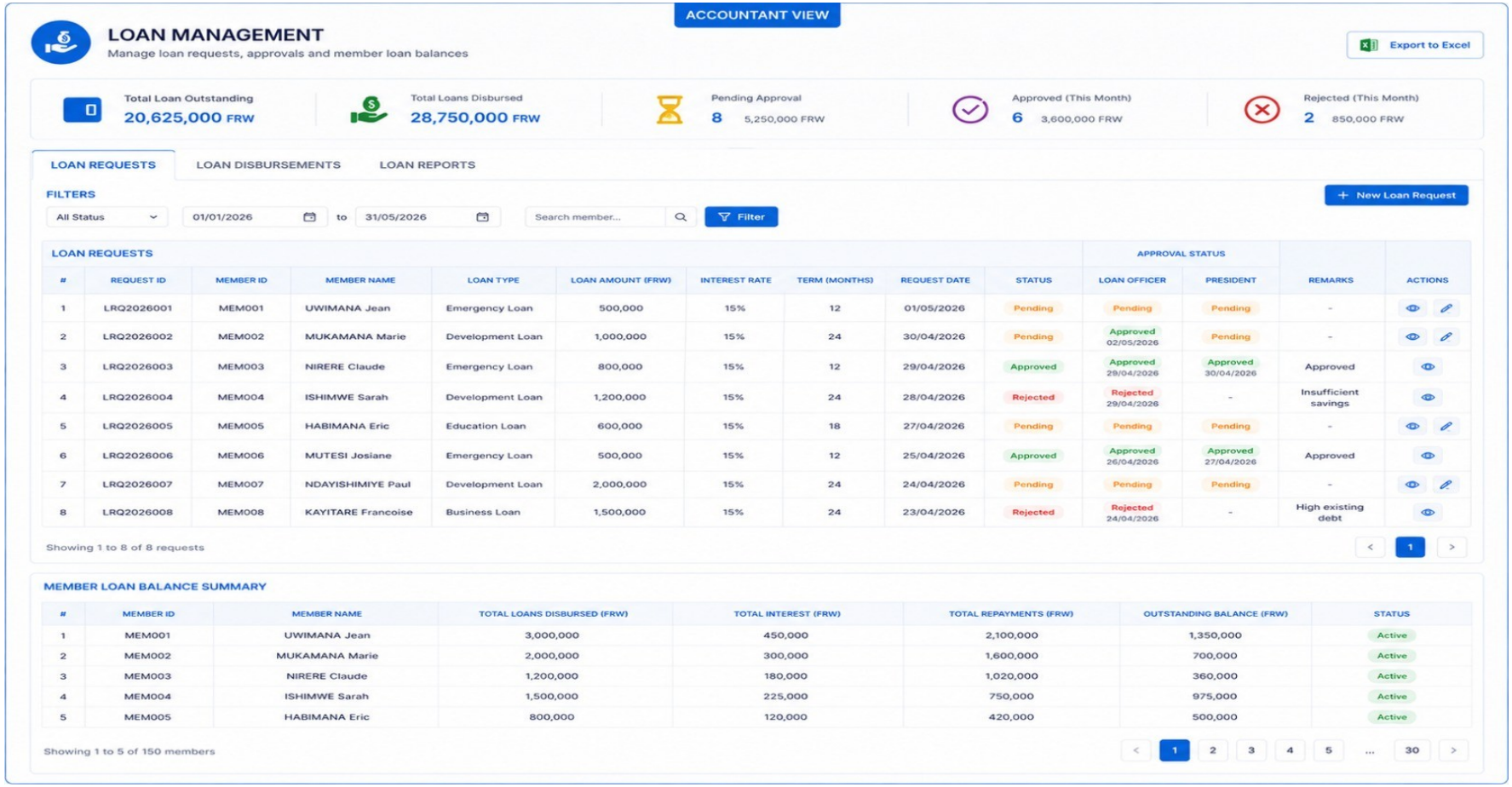

USER VIEW
SAVINGS DETAILS
Review your savings and contribution history
Initial Capital (Non-refundable) 50,000 FRW
Cumulative Savings 125,750 FRW
Cumulative Interest 15,250 FRW
Total Savings 191,000 FRW
Monthly Contribution 10,000 FRW
SAVINGS OVERVIEW
PERIOD | DATE | DESCRIPTION | MONTHLY CONTRIBUTION (FRW) | INTEREST (FRW) | TOTAL BALANCE (FRW)
0 (Initial Capital) | 01/01/2026 | Initial Capital (Non-refundable) | - | - | 50,000
1 | 01/01/2026 | January 2026 contribution | 10,000 | 250 | 60,250
2 | 01/02/2026 | February 2026 contribution | 10,000 | 260 | 70,510
3 | 01/03/2026 | March 2026 contribution | 10,000 | 270 | 80,780
4 | 01/04/2026 | April 2026 contribution | 10,000 | 280 | 91,060
5 | 01/05/2026 | May 2026 contribution | 10,000 | 290 | 101,350
Interest is calculated monthly on your savings balance.
CHANGE MONTHLY CONTRIBUTION
Update your monthly savings contribution amount.
Current Monthly Contribution
10,000 FRW
New Monthly Contribution
12,000 FRW
Update Contribution
HISTORICAL CONTRIBUTION REPORT
Export to Excel
MONTH | DATE | MONTHLY CONTRIBUTION (FRW) | INTEREST (FRW) | TOTAL DEPOSIT (FRW) | BALANCE (FRW)
January 2026 | 01/01/2026 | 10,000 | 250 | 10,250 | 60,250
February 2026 | 01/02/2026 | 10,000 | 260 | 10,260 | 70,510
March 2026 | 01/03/2026 | 10,000 | 270 | 10,270 | 80,780


ACCOUNTANT VIEW
SAVINGS MANAGEMENT
Manage member savings, contributions, and withdrawals
Export to Excel
FILTERS
All Members
01/01/2026
to
31/05/2026
Filter
Total Members 150
Total Savings 18,750,000 FRW
Total Interest 1,875,000 FRW
Total Savings Balance 20,625,000 FRW
MEMBER MANAGEMENT
CONTRIBUTIONS
WITHDRAWALS
REPORTS
Manage all members and their savings
Add New Member
Add Contributions for All
Search members...
Download Template
| MEMBER ID | MEMBER NAME | INITIAL CAPITAL (FRW) | MONTHLY CONTRIBUTION (FRW) | TOTAL CONTRIBUTIONS (FRW) | INTEREST (FRW) | TOTAL SAVINGS (FRW) | STATUS | ACTIONS
1 | MEM001 | UWIMANA Jean | 50,000 | 10,000 | 120,000 | 12,000 | 182,000 | Active
2 | MEM002 | MUKAMANA Marie | 50,000 | 15,000 | 180,000 | 18,500 | 248,500 | Active
3 | MEM003 | NIRERE Claude | 50,000 | 20,000 | 240,000 | 24,000 | 314,000 | Active
4 | MEM004 | ISHIMWE Sarah | 50,000 | 10,000 | 110,000 | 10,800 | 170,800 | Active
5 | MEM005 | HABIMANA Eric | 50,000 | 15,000 | 150,000 | 15,000 | 215,000 | Active
Showing 1 to 5 of 150 members
1 2 3 4 5 ... 30

MODULE 3: LOAN MANAGEMENT

User module example:





7.1 Loan Application

- Members fills in the loan request form
- System calculates eligibility automatically
- Enter apply for loan and enter minimum data and upload signed form

7.2 Loan Eligibility Formula

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Loan Limit = (Monthly Saving / 3 × 24) + Total Saving
```

7.3 Loan Rules

- Maximum Loan: 5,000,000 FRW
- Interest Rate: 5% (simple interest)
- Period: Max 24 months

7.4 Loan Workflow

1. Application submission
2. Loan Officer review
3. President approval
4. Disbursement
5. Monthly repayment tracking

7.5 Loan Tracking Fields

Field	Description
Loan Amount	Approved amount
Interest	Calculated
Total Payable	Principal + Interest
Monthly Installment	Calculated
Balance	Remaining loan

MODULE 4: FUND ADMINISTRATION

8.1 Features

- Transaction approvals
- Audit logs
- Financial reporting

8.2 Approval Workflow

- Dual authorization required:
  - Accountant (entry)
  - President / Loan Officer (approval)

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6. INTEREST CALCULATION

Savings Interest

- Distributed proportionally based on member savings

Loan Interest

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Interest = Principal × Rate × Time
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7. REPORTS

7.1 Standard Reports

- Member Savings Statement
- Loan Statement
- Social Fund Cash Book

- Monthly Contributions Report
- Interest Distribution Report

7.2 Financial Reports

- Monthly reconciliation
- Quarterly financial report
- Annual summary

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8. SPECIAL FEATURES

8.1 Loan Offset

- Member can request:
    - Offset loan balance with savings
    - Reset account

8.2 Payroll Integration

- Import monthly contributions
- Auto-post entries
- Of manual excel upload

8.3 Notifications

- Loan approvals
- Remembers of payment
- Contribution alerts

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9. DATABASE DESIGN (HIGH LEVEL)

Main Tables

- Users
- Members
- Savings
- Loans
- Loan Repayments
- Social Fund Transactions
- Approvals
- Reports

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10. SECURITY REQUIREMENTS

- Password encryption
- Role-based access
- Audit trails
- Backup & recovery system

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11. TECHNOLOGY RECOMMENDATION

Frontend

- Web-based (React / Angular)

Backend

- Node.js / Django / Laravel

Database

- PostgreSQL / MySQL

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12. FUTURE ENHANCEMENTS

- Mobile App
- SMS notifications
- Integration with banking systems
- Biometric login

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13. SUCCESS CRITERIA

- Accurate tracking of funds
- Transparency in operations
- Fast loan processing=
- Easy reporting and auditing